Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
YEAR 6

HIGHER 2 GEOGRAPHY **9173/01**

Wednesday **27 September 2024** **3 hour**

Indicative Content

Cluster 1: Development, Economy and Environment

1(a) Using an example for each, explain what is meant by conditional reserves and hypothetical reserves. [4]

Award 1 mark for each explanation using an example of what is meant by conditional reserves and hypothetical reserves.
Award a maximum of 1 additional mark for further development of each explanation. Award a maximum of 2 marks for explanation of <u>either</u> conditional <u>or</u> hypothetical reserves only.

Possible responses include:
*- Oil sands in Canada can be considered **conditional** reserves because they have been discovered but face limitations in being extracted. [1 mark]*
- They require a minimum of the oil prices to be around $65. When price falls below this level, they become economically unviable to extract. [1 additional mark]

1(b) Compare the distribution of palm oil production between 1992 and 2022 shown in Resource 1. [4]

Award 1 mark for each comparison.
Candidates are expected to compare the distribution using the proportional symbols (circles) to give similarities and differences.
Candidates are not expected to know individual countries but may refer to them in their response.

Possible responses include:

Similarities
- The number of continents (4) where palm oil production can be found is the same in 1992 and 2002 – North America, South America, Asia and Africa. [1 mark]
- The countries producing palm oil in 1992 and 2002 are all close to the equator/within the tropics. [1 mark]
- Asia is the region with the largest production in both 1992 and 2002. In 1992, it is the only region which had countries with more than 2 million MT; in 2002, it remains the only region which had countries with more than 2 million MT. [1 mark]
Differences
- The number of countries where palm oil production can be found is higher in 2022, which saw 26, than in 1992, which had 24. [1 mark]

- North America, South America and Asia saw an increase in production sites (+3) between 1992 and 2022 whereas Africa saw a decrease in production sites (-1).

1(c) With reference to Resource 2, explain how palm oil production influences progress towards sustainable development in Indonesia and Malaysia. [6]

Award 1 mark for an explanation of how palm oil production promotes progress towards SD
Award 1 mark for an explanation of how palm oil production hinders progress towards SD
Award a maximum of 1 additional mark for further development of each explanation

Possible responses include:

<u>Promotes sustainable development by:</u>
*- Significantly contributing to Malaysia and Indonesia's national **economic growth**, one way to progress towards sustainable development, as seen from the contribution of this industry to both countries' Gross Domestic Product [1 mark]*

<u>Hinders sustainable development by:</u>
*- Compromising **environmental protection** when huge expanse of forests are being cleared to make way for palm oil plantation, causing harm to natural ecosystems [1 mark] This can retard progress towards the attainment of **Sustainable Development Goal 15 Life on Land**. [1 additional mark]*

1(d) Using Resource 3, contrast the solar and hydroelectric generation in USA from 2019 to 2024. [4]

Award 1 mark for each difference outlined.
Award 1 additional mark for corresponding evidence to show difference.

Possible responses include:

*- **Difference in annual pattern:** Solar generation has a consistent pattern whereas hydroelectric generation has a fluctuating pattern. [1 mark]*
*- **Difference in trend over time:** solar generation shows an increasing trend over time whereas hydroelectric generation shows a relatively stable trend. [1 mark]*

1 (e) Suggest why monthly solar generation in USA is likely to increase in the future as shown in Resource 3. [6]

Award 1 mark for each explanation.
Award a maximum of 1 additional mark for further development of each explanation.

Possible responses include:

- Because of its reliability (the Sun will continue to exist for billions of years), it will be chosen over other renewable energy sources such as hydropower, whose reliability may be compromised by changing weather patterns due to climate change. [1 mark]
- Over the years, the cost of solar panel technology has significantly decreased, making solar power increasingly cost-competitive with fossil fuels. [1 mark]
- There could be growing awareness of environmental issues in a developed country like USA and the urgency of renewable energy to mitigate climate change drives greater adoption of solar power. [1 mark]

1(f) Explain how the schematic diagram in Resource 4 shows Thomas Malthus' views on population and resources. [6]

Award 1 mark for each explanation.

Possible responses include:

- Malthus believes that over time, the population will exceed the ability of the environment to support it. [1 mark] R4 shows that the increase in population, through increased demand for energy resources, eventually led to food shortage. [1 mark]

Cluster 2: Tropical Environments

2 Resource 5 shows the position of the ITCZ and wind systems in the tropics in December and July. Resource 6 shows the relief map of Indian sub-continent, and the rainfall received in July at locations A, B and C. Resource 7 shows a developed river channel of the Mekong River that is in flood in Cambodia. Resource 8 is a schematic diagram of the stages in the development of the river channel in Resource 7.

(a) With reference to Resource 5, describe the differences in the features of the ITCZ and the wind patterns in December and July [5]

Award 1m for description of each difference
Award a maximum of additional 1m for further development of each description where applicable

Indicative content
The differences can focus on the characteristics ie the position/location and the pattern of the ITCZ and direction of the wind systems at the Asia landmass and at the ITCZ

ITCZ:

- ***Position/ Location**: located just north of the Tropic of Capricorn in Jan but slightly both north and south of the Tropic of Cancer in July **[1 mark]** and predominantly over the oceans – Indian and Pacific oceans in Jan but over the continents - Asian landmass in July **[1 additional mark]***

Wind systems in Asia (NH)
Direction**: reversal of prevailing winds from Jan to July with the north-east winds changing to south-west winds **[1 mark]

(b) With reference to Resources 5 and 6, explain the difference in the rainfall received in July for stations A, B and C in Resource 6. [6]

Award maximum of 2m for explanation of prevailing onshore monsoon wind characteristics & orographic influence'
Award 1m for explanation of rainfall at each station to a maximum of 5m
Award maximum of 1 additional mark for explanation where applicable
Award maximum of 2m for explanation of just one station or maximum of 4m for just 2 stations
Indicative content
Response should have understanding of prevailing onshore winds at western coast, moisture supply from the Indian Ocean, air instability and uplift eg:

Proximity to onshore winds on the west coast

- *A is closest to the prevailing onshore winds on the west coast– only 80 km from the coast, B about 350 km away but C is the furthest inland of 600 km so winds will be most moist at A and be driest by the time the winds reach C*

Extent of air mass uplift

- *A is located on the windward side of the coastal mountains where air uplift occurs and receives heavy rainfall but both B and C experience the leeward effect where descent of air masses is experienced and will receive less rain.*

(c) With reference to Resource 8, compare the river channel at Stage 1 with the river channel at Stage 3. [4]

Award 1m for each description of similarity and difference
Award a maximum of 3m for description of either similarity or difference

Indicative content
The comparison can focus on the channel characteristics eg channel pattern, length, sinuosity, multiplicity and the areas of erosion and deposition eg.

Differences:

Areas of deposition:
***Mid-channel**: Stage 3 has 6 areas of deposition, but Stage 1 has only 4 – Stage 3 has 2 more areas of deposition than Stage 1 [**1 mark**]*
***Side of channel**: Stage 3 has smaller and more areas of deposition (7) but Stage 1 has fewer (3) but larger areas of deposition [**1 mark**]*

Areas of erosion
*Stage 3 has eroded banks but none in Stage 1 [**1 mark**]*

Similarities
Areas of deposition
*Both stages have areas of deposition in the middle and at the side of the channel [**1 mark**]*

(d) With reference to Resource 8, explain how the fluvial processes operating in the channel in Stages 2 and 3 will result in the developed channel in Resource 7. [7]

Award 1m for each stage of the processes to a maximum of 6m
Award maximum of 2m for well-annotated diagram of the processes

Indicative content
The content should feature an understanding of the channel bedforms to initiate the deflection of the primary flow and the generation of the secondary helicoidal flow across the channel. The erosion processes at the outer/concave bank should be developed to explain the formation of the river cliff and how the eroded sediments are transported by the helicoidal flow across the channel for deposition to occur at the inner/convex bank to build up a shallow bed of deposition over time to form the point bar. Continual lateral erosion and deposition over time will increase channel sinuosity to develop the channel shown in R7.

(e) Name and describe the feature labelled 'X' and the feature between Y and Z in Resource 7. [4]

Award 1m for name of each feature
Award 1m for description of feature
Award a maximum of 2m for name and description of either feature

Indicative content
*X: point bar [1 mark]: a large/wide, gently sloping / shelving area of deposition that is elevated above the water level and located on the inner bend of a meander [**1 mark**]*

*Y: river cliff [1 mark]: a narrow steeply sloping bank that is constantly being eroded /undercut at the base and located on the outer bank [**1 mark**]*

(f) With reference to Resource 7, explain how the nature of the channel and the physical processes operating in the channel may affect Settlements A and B differently. [4]

Indicative content
Nature of channel
Settlement A

- *gently shelving bank - R3 shows flooding at the inner bend of the meander as flow velocities are slow and the floodwaters do not drain away easily due to the gentle gradient – settlement A will be affected by the floods.*

Physical process
Settlement B
Settlement B is located at the bank where the fast water currents are and slope undercutting occurs easily and the top of the slope will be unstable and may collapse due to slope processes – houses at Settlement B will be badly affected by slope collapse.

Cluster 1 Development, Economy, Environment SEQs

3 'It is challenging to manage conflicts over transboundary water resources.'

Evaluate this statement. [20]

Indicative Content

Conflicts over transboundary water resources, defined as disagreements among riparian states over the use and management of shared water bodies, are increasingly challenging to manage. These conflicts arise from competing needs for water for agriculture, hydropower, and industry, all of which are exacerbated by drivers such as population growth and economic development. Despite various efforts to mitigate these conflicts, management of transboundary water disputes remains a daunting task. Challenges come in the form of strategic challenges where the nature and scale of strategies implemented are ineffective to manage conflicts, as well as the economic and demographic contexts of the riparian states where increased water demand leads to increased tension. The scale of empirical examples worldwide that show unresolved conflicts is significant and makes this statement highly valid. However, there exists a handful of instances where the challenges to manage conflicts have been overcome.

4 'The GPNs of TNCs have resulted in greater interdependence between places within the global economy.'

Evaluate this statement. [20[

Indicative content

"Interdependence" can be understood as a relationship in which two places rely on each other to meet their needs or function effectively within the global economy. The idea of interdependence can be clearly seen in the historical relationship between developed and developing regions. Developing macro-regions, such as Sub-Saharan Africa, often rely on the export of minerals like gold and cobalt to countries in developed regions. In return, these developing regions depend on developed countries for importing essential goods like machinery, technology, and pharmaceuticals. Since the 1950s, the Global Production Networks (GPNs) of Transnational Corporations (TNCs) have significantly reshaped the way places are connected within the global economy. By integrating different macro-regions through complex webs of production, TNCs have played a pivotal role in fostering ***even greater*** *interdependence between places. As a result, the GPNs of TNCs have resulted in greater interdependence in terms of flows of* ***material*** *and* ***capital*** *between most places within the global economy.*

Cluster 2 Tropical Environments SEQs

5 Evaluate the role of geomorphic processes in the formation of tropical karst landforms. [20]

Indicative content

The range of geomorphic processes that form the tropical karst are weathering, slope and fluvial processes. To evaluate the role of these processes, would need to examine the extent of involvement of each of these processes in the formation of the karst landforms. For each geomorphic process, need to consider the 'scale' of the involvement eg the frequency of the process and spatial scale of operation. The extent of the involvement will be dependent on the environmental conditions / factors. The landforms to be considered should showcase the range of geomorphic processes. The response should be organized according to geomorphic processes and not by the environmental factors.

6 'Humans cannot control the occurrence of fluvial floods' Evaluate this statement. [20]

Indicative content

Candidates could approach the question by making a judgement on whether fluvial floods can be controlled by humans through a consideration of the factors contributing to the occurrence of floods – the timing, frequency as well as flood magnitude at various scales. Broadly speaking, humans would not be able to control flood occurrence as this is largely determined by the factors that operate at the macro (climate) and meso- scales (basin). However, at the local scale, humans would have a larger control over flood magnitudes through activities that modify the basin and channel characteristics. However, candidates should ***evaluate across space*** *how the extent of human control may vary due to difference in contextual factors such as the nature of the flood and preparedness of the population. But, it should be noted that humans have little control over flood occurrence when mitigation measures would be ineffective in large drainage basins or entire rivers and against more intense rainfall events and storm activity in the tropics due to worsening climate change over time. In a consistently evaluative response, candidates could make a judgment on whether fluvial floods can be controlled by humans through a consideration of two or more case studies. Candidates could also evaluate, through appropriate criteria, whether the factors are within the control of humans.*